



## Definition of Z sub alpha

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In my experience,  $Z_\alpha$  indicates the critical value where the right-tailed area under a standard normal distribution is  $\alpha$ , i.e.

$$P(Z > Z_\alpha) = \alpha$$



With this rule, if  $\alpha = 0.05$ , then

$$Z_{\frac{\alpha}{2}} = Z_{0.025} = 1.96 \quad \text{and} \quad Z_{1-\frac{\alpha}{2}} = Z_{0.975} = -1.96.$$



However, in some internet sources,  $Z_\alpha$  is defined as the inverse function of the standard normal CDF, i.e.

$$P(Z < Z_\alpha) = \alpha$$

which makes the example above have opposite signs:

$$Z_{\frac{\alpha}{2}} = -1.96 \quad \text{and} \quad Z_{1-\frac{\alpha}{2}} = 1.96.$$

They make me confused about how to use correct symbols in the confidence intervals or test statistics. Is there a precise mathematical definition of  $Z_\alpha$ ?

[hypothesis-testing](#) [confidence-interval](#) [inference](#)

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edited Nov 2, 2020 at 20:22

asked Nov 2, 2020 at 20:17

 **darren**  
41   1   3

4  Clearly both definitions are precise and mathematical. It's a matter of convention. Since the most popular convention in statistics is to define the distribution function in terms of  $\Pr(Z \leq z)$ , the latter meaning is usually what is intended. – [whuber](#) ♦ Nov 2, 2020 at 20:53

### 1 Answer

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No it is not correct to say that  $P(Z < z_\alpha) = \alpha$ . It is  $P(Z < z_\alpha) = 1 - \alpha$  which is equivalent to  $P(Z > z_\alpha) = \alpha$ .



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answered Jan 9, 2022 at 11:38

 **Ahad Dehghani**  
11   1



5  Both are correct, depending on how  $Z_\alpha$  is defined. It would be a mistake to assume the whole world has adopted *your* convention. – [whuber](#) ♦ Jan 9, 2022 at 19:36